

# GUHNOO YUN

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## RESEARCH STATEMENT

**Spectral analysis and adaptive filtering for visual perception, from recognition to image generation.** My research is about visual perception for machine intelligence: I analyze what vision models pass or suppress in the spectral domain and turn that analysis into operators for visual recognition and image generation.

- Spectral analysis of convolution and self-attention (frequency balancing, graph-spectral frequency responses)
- Spectral-adaptive operators as general-purpose replacements for softmax attention, from vision backbones to diffusion transformers (SPAM, SALA)
- Visual perception for robots and healthcare (fall detection, isolation-ward services, daily-life monitoring)

In the next step, I plan to do the same for video generation and world models at a larger scale.

## EDUCATION

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| Feb 2027<br>(expected) | <b>Ph.D. in Computer Science and Engineering</b><br><b>Korea University</b> Seoul, Korea<br>Dissertation: <i>Spectral Analysis and Adaptive Filtering for Visual Perception</i>                                  |
| 2018                   | <b>M.S. in Mechatronics</b><br><b>Gwangju Institute of Science and Technology (GIST)</b> Gwangju, Korea<br>Thesis: <i>Removal of Jitter Noise using a Modified Kalman Filter in 3D Shape Recovery from Focus</i> |
| 2016                   | <b>B.S. in Control and Measurement Engineering</b><br><b>Gyeongsang National University</b> Jinju, Korea                                                                                                         |

## RESEARCH EXPERIENCE

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| Since 2019 | <b>Student Researcher</b><br>Intelligence and Interaction Research Center, <b>Korea Institute of Science and Technology (KIST)</b><br>Supervisor: Dr. Dong Hwan Kim<br><b>Linear Attention for Generative and Vision Transformers (SALAFormer; in preparation [11])</b> <ul style="list-style-type: none"><li>• Designed <b>SALA</b> (Spectral-Adaptive Linear Attention) as closed-form associative memory: each head solves a small ridge regression for what to store in its fixed-size memory, instead of summing raw key-value products, and a learned spectral filter on the key Gram matrix allocates that memory, at linear <math>O(N)</math> inference cost.</li><li>• <b>Image generation:</b> lower FID than MHLA (ICLR 2026), a recent linear-attention design for diffusion transformers, at both model sizes: 59.80 <math>\rightarrow</math> <b>57.65</b> (DiT-S/2) and 37.47 <math>\rightarrow</math> <b>36.54</b> (DiT-B/2) vs. its published results (class-conditional ImageNet 256<math>\times</math>256, no guidance).</li><li>• <b>Image classification:</b> SALA in a vision backbone (SALAFormer) reaches <b>77.2%</b></li></ul> |
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/ **81.8%** ImageNet-1K top-1 at DeiT-Tiny/Small size, above the 76.5% / 81.6% reported for ViT<sup>3</sup> (CVPR 2026) with the same recipe.

### Spectral Analysis of Convolution and Attention for Vision Backbones

- **SPANetV2** (TPAMI 2026): compared what convolution and self-attention pass or suppress in one graph-spectral framework, showing that window size shapes their frequency responses, and turned this into a mixer with multi-scale kernels and spectral re-scaling; the backbone outperforms state-of-the-art models on ImageNet-1K, COCO, and ADE20K. [1]
- **SPANet** (ICCV 2023): prior work found that self-attention acts as a low-pass filter, while convolution retains high frequencies, and that stronger high-pass filtering helps self-attention. SPANet showed that stronger low-pass filtering also helps convolution, and that a token mixer *balancing* high and low frequencies improves accuracy. [4]

### Applied Vision for Robotics, Healthcare & 3D Shape Recovery

- Fall detection of older adults by a healthcare robot [6]; robot-assisted services in isolation wards [5]; daily-life arousal monitoring and brain-computer interface decoding with egocentric vision and EEG [2, 3]; a view-consistent evaluation metric for multi-view panoptic segmentation [10].
- Shape from focus, a follow-up to the M.S. thesis: 3D shape from few images with Gaussian-process regression [7].

2018 – 2019 **Intern Researcher**

Center for Intelligent Robotics, **KIST**

- Perception for healthcare robots: detecting when people with dementia leave home, to keep them from going missing [9]; accelerometer–depth fall detection for older adults [8], later extended with LSTMs [6].

## PUBLICATIONS

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### Published / Accepted

- [1] Guhnoo Yun, J. Yoo, K. Kim, J. Lee, P.H. Seo, D.H. Kim, **Spectral-Adaptive Modulation Networks for Visual Perception**. *IEEE Trans. Pattern Anal. Mach. Intell. (TPAMI)* 48(10), 11511–11525 (2026). [doi: [10.1109/TPAMI.2026.3690455](https://doi.org/10.1109/TPAMI.2026.3690455)] [arXiv] [code]
- [2] W. Hyung, M. Kim, Y.-S. Kim, J. Lee, B. Ko, Guhnoo Yun, D.H. Kim, C.-H. Im, **Context-Aware Daily-Life Arousal Monitoring Using Egocentric Vision and EEG**. *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)* (2026). [program]
- [3] Y.-S. Kim, M. Kim, W. Hyung, B. Ko, Guhnoo Yun, D.H. Kim, C.-H. Im, **PeBCI: Vision-Guided Brain-Computer Interface for Context-Aware Neural Decoding in Daily Life**. *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)* (2026). [program]
- [4] Guhnoo Yun, J. Yoo, K. Kim, J. Lee, D.H. Kim, **SPANet: Frequency-balancing Token Mixer using Spectral Pooling Aggregation Modulation**. *IEEE/CVF Int. Conf. on Computer Vision (ICCV)*, 6090–6101 (2023). [doi: [10.1109/ICCV51070.2023.00562](https://doi.org/10.1109/ICCV51070.2023.00562)] [CVF] [arXiv] [project] [code]
- [5] Y. Kwon, S. Shin, K. Yang, S. Park, S. Shin, H. Jeon, K. Kim, Guhnoo Yun, ..., Y. Lim, **Heterogeneous Robot-Assisted Services in Isolation Wards: A System Development and Usability Study**. *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 8069–8076 (2023). [doi: [10.1109/IROS55552.2023.10341857](https://doi.org/10.1109/IROS55552.2023.10341857)]
- [6] K. Kim, Guhnoo Yun, S.-K. Park, D.H. Kim, **Efficient Fall Detection for a Healthcare Robot System Based on 3-Axis Accelerometer and Depth Sensor Fusion with LSTM Networks**. *Int. Conf. on Pattern Recognition (ICPR)*, 2207–2212 (2022). [doi: [10.1109/ICPR56361.2022.9956418](https://doi.org/10.1109/ICPR56361.2022.9956418)]

- [7] H.-S. Jang, Guhnoo Yun, M.T. Mahmood, M.-K. Kang, **Optimal Sampling for Shape from Focus by Using Gaussian Process Regression**. *IEEE Int. Conf. on Consumer Electronics (ICCE)*, 1–4 (2020). [doi: [10.1109/ICCE46568.2020.9043150](https://doi.org/10.1109/ICCE46568.2020.9043150)]
- [8] K. Kim, Guhnoo Yun, S.-K. Park, D.H. Kim, **Fall Detection for the Elderly Based on 3-Axis Accelerometer and Depth Sensor Fusion with Random Forest Classifier**. *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, 4611–4614 (2019). [doi: [10.1109/EMBC.2019.8856698](https://doi.org/10.1109/EMBC.2019.8856698)]
- [9] Guhnoo Yun, K. Kim, S.-K. Park, D.H. Kim, **A Monitoring System to Support Home Health Care for the Elderly with Dementia by Detecting Going Out Activities Based on RGB-D Sensors**. *Int. Conf. on Ubiquitous Robots (UR)*, 71–76 (2019). [doi: [10.1109/URAI.2019.8768598](https://doi.org/10.1109/URAI.2019.8768598)]

#### Under Review

- [10] Y. Lee, B. Ko, Guhnoo Yun, D.H. Kim, **Every View Counts: View-Consistent Panoptic Quality for Multi-view Panoptic Segmentation**. *arXiv preprint* (2026). [arXiv: [2610.05911](https://arxiv.org/abs/2610.05911)]

#### In Preparation

- [11] Guhnoo Yun et al., **SALAFormer: Spectral-Adaptive Linear Attention as Closed-Form Associative Memory**.

### AWARDS & SCHOLARSHIPS

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- 2026 **Outstanding Student Researcher Scholarship**  
Korea Institute of Science and Technology (KIST)
- 2023 **Outstanding Technology Award**  
Korea Institute of Science and Technology (KIST)
- 2020 **Presidential Award** 2020 International Robot Contest—Humanoid Robot Sports (Deep Learning Algorithm)  
Ministry of the Interior and Safety

### INVITED TALKS

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- 2021 **“Recent Research Trends in Computer Vision”**  
Seminar at Korea Polytechnic University (now Tech University of Korea)
- 2021 **“6D Object Pose Estimation”**  
Seminar series at CUBOX (now SECERN AI)

### PROFESSIONAL SERVICE

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- 2024, 2026 **Reviewer, International Conference on Pattern Recognition (ICPR)**

### TECHNICAL SKILLS

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**Languages & Frameworks:** Python, PyTorch, C++, MATLAB; timm, MMDetection, MMSegmentation, ROS

### REFERENCES

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**Dr. Dong Hwan Kim**  
KIST  
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